Finvexis AI: Agentic Business Intelligence Platform

Introduction toGenerativeArtificial Intelligence|CSET-419

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The Problem & Our Solution

The Problem

- Businesses drown in data across Finance, HR, and Sales daily
- Static dashboards and manual reports can't deliver real-time insights
- Existing tools are siloed — no cross-departmental intelligence
- Heavy reliance on human resources for routine analysis

Our Solution — Finvexis AI

- Multi-agent, modular Business Intelligence platform
- Automates Financial Analysis, HR Operations, Sales Intelligence & Strategic Reporting
- Natural Language interface — no technical background needed
- Each department runs as an independent AI-powered module

How It All Connects

01

Frontend

React / Next.js captures user input

02

Gateway

FastAPI routes requests to the correct module

03

Module Pattern

Orchestrator → Agents → Data Layer → Response

04

Dep ar tments

Finance · Sales & HR · Business Intelligence

05

API Gateway

All connected through a single API gateway

Finance Department — 4 Specialized Agents

Strict sequential execution enforced: **Invoice→ Auditor →VendorIntel →FP&A**

All risk scoring, tax math, and budget logic runs on **pure deterministic Python — no LLM**

LLM is only used to narrate pre-computed results

Agent	What It Does
Invoice Agent	PDF parsing, GST/TDS detection, basic risk flagging
Auditor Agent	Duplicate detection, missing GST, spending spike alerts
Vendor Intelligence	Risk scoring (0–100), rate hike & TDS gaming detection
FP&A Agent	Budget vs actuals, burn rate tracking, what-if simulations

Sales & HR — NLP-Powered Operations

1 Quer y

2 Intent Classification

3 Agent

4 Response

CRM Layer
SQLite database with full customer interaction history

Lead Scoring
Weighted score based on budget, company size, decision-maker engagement

Sales Advisor
Personalized outreach drafts using live CRM context

HR Policy Bot (RAG)
Answers employee queries strictly from Vector DB — zero hallucination

Resume Engine
Deterministic match score (0–100) comparing candidate skills vs job description

BI Department — Enterprise-Level Intelligence

Operates at macro level using internal CSVs + external market data

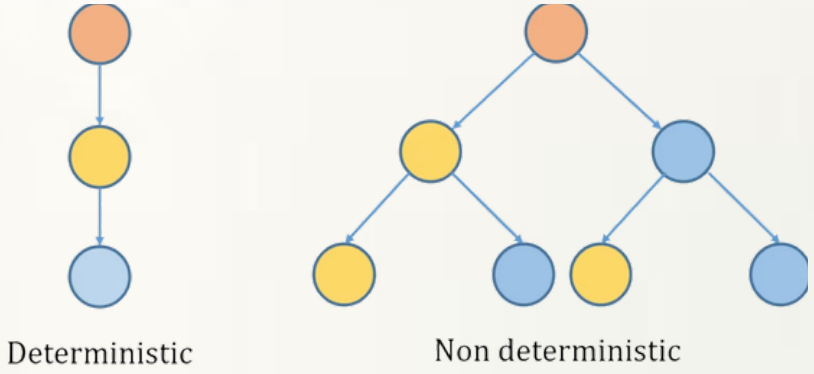
Parallel Phase

KPI Agent + Forecasting Agent run concurrently

Sequential Phase

Market Agent → Strategy Agent synthesizes everything

Agent	Role
KPI Agent	Revenue growth, margins, Z-score anomaly detection, composite health score
Forecasting Agent	12-month linear regression — Realistic / Best (+15%) / Worst (−15%) scenarios
Market Agent	Pricing index, market positioning score, opportunity scoring
Strategy Agent	LLM-powered strategic roadmap with deterministic fallbacks



Deterministic Core + Constrained LLM

Rule
Every financial decision, risk score, and budget calculation is pure Python

LLM Role
Only narrates what the deterministic engine has already computed

Why This Matters

- Fully auditable — every output can be traced to a formula
- Testable — each agent works independently
- Reliable — LLM can be swapped without touching business logic
- No hallucination risk in financial or compliance decisions

Key Features in Action

Finance Module

- Invoice upload → automatic GST routing (Intra/Inter-state) + TDS detection
- Auditor flags duplicate invoice & 3 months of missing GST from same vendor
- FP&A dashboard shows IT over budget by ~45%, Marketing under at ~75%

Sales & HR Module

- CRM logs a new lead and auto-scores it based on budget, size & engagement
- Sales Advisor drafts a personalized follow-up email using live CRM data
- Resume Engine scores a candidate (0–100) against a job description instantly
- HR Policy Bot answers employee queries directly from company policy documents

Business Intelligence Module

- KPI Agent detects a statistical anomaly using rolling Z-score analysis
- Forecasting Agent projects 12-month revenue in 3 scenarios — Realistic, Best, Worst
- Market Agent benchmarks company KPIs against industry competitors
- Strategy Agent outputs a full multi-timeframe strategic roadmap

Where We Are & Where We're Going

Current Limitations

- Data is not persistent — all modules reset on app restart, no production DB yet
- All input is static — Finance uses hardcoded Q1 2026 data, BI relies on CSV uploads, CRM runs on local SQLite
- No live data feeds — market benchmarking, policy documents, and budgets are all hardcoded
- Scope constraints — Finance supports only flat 18% GST, Forecasting uses only Linear Regression, Resume Engine accepts PDF only
- No API authentication or role-based access control — not production-ready for external deployment

Future Roadmap

- Migrate to PostgreSQL + Pinecone for scalable persistent storage
- Containerize into microservices managed by Kubernetes + Apache Kafka
- Live ERP/CRM integrations with SAP and Salesforce
- Dynamic model routing — frontier models for complex tasks, local models for quick queries

Key Takeaways

Logic & Narration Separated
Finvexis AI proves that multi-agent BI works best when logic and narration are cleanly separated

Auditable & Trustworthy
Deterministic engines make every financial decision auditable and trustworthy

Easy Expansion
Modular architecture allows easy expansion to new departments without breaking existing ones

Scales Across Functions
The same pattern — deterministic core + constrained LLM — scales across all business functions

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Thank you!

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